

AgriShield

Technology, AI Models and Feature Report

Clear project status - generated 09 September 2026

About this report. This document describes what the local project currently implements. It separates working features, trained artifacts and future work so that the presentation stays accurate.

1. Features added to AgriShield

- **Leaf disease assistance:** image upload, phone/PC camera capture, and middle-frame extraction from a short leaf video. The app returns an advisory class, confidence, guidance and medicine options.
- **No-image symptom check:** farmers can type visible symptoms and receive a possible risk and next step. This is not a confirmed diagnosis.
- **Live weather:** OpenWeather current conditions and 5-day forecast, with humidity/rainfall risk notices. It needs the server environment variable `OPENWEATHER_API_KEY`.
- **Farm tools:** soil health guidance, crop recommendations, yield simulation, irrigation planner and fertilizer nutrient-gap calculator.
- **Farmer and doctor workflow:** login/register, plants, scan history, appointment booking, doctor profiles, chat/voice room, medicine orders and dashboard.
- **Offline assistant:** saved guidance is available from the service-worker cache; server features reconnect when online.

2. Technology stack

- **Frontend:** HTML, CSS, vanilla JavaScript, Progressive Web App service worker, browser camera APIs, Canvas image capture and Web Speech APIs.
- **Backend:** Python, FastAPI, Uvicorn, SQLAlchemy and SQLite.
- **Security:** Passlib bcrypt password hashing and JWT bearer tokens.
- **Image and machine learning:** Pillow, NumPy, scikit-learn RandomForestClassifier. The separate training environment uses TensorFlow/Keras and MobileNetV2.
- **External service:** OpenWeather API, called by the server so the API key is not exposed in the browser.

3. Algorithms currently used

- **Live leaf predictor:** a lightweight 128 x 128 RGB visual-feature pipeline measures green, yellow, brown, dark and texture signals and maps them to five advisory classes.
- **Environmental risk:** an 80-tree Random Forest predicts low, medium or high risk from crop, month, rainfall, temperature, humidity and prior outbreak.
- **Symptom triage:** rule matching identifies patterns compatible with fungal, bacterial, nutrient stress or pest damage.
- **Farm calculations:** transparent scoring and benchmarks generate crop suitability, yield, irrigation and nutrient-gap estimates.

4. CNN included in the project

A trained Keras artifact is stored as **models/rice_leaf_disease.keras**. It was trained using MobileNetV2 transfer learning for three rice-leaf classes: Bacterial leaf blight, Brown spot and Leaf smut.

- **Training data:** 120 rice-leaf images, recorded as 40 images per class.
- **Training run:** 12 epochs; recorded training accuracy 81.25 percent; reported validation accuracy 75.00 percent.
- **Architecture:** MobileNetV2 feature extractor plus classification head, image augmentation, sparse categorical cross-entropy and Adam optimization.
- **Current application status:** the live /api/predict path still uses the lightweight visual-feature fallback. The saved CNN is available for integration after TensorFlow is installed in the serving environment and validation is corrected.

Important validation warning. The saved validation report has only one class with validation support and a malformed confusion matrix. The reported 75 percent validation score must not be used as a claim of model quality across all three classes. Retrain using a stratified train, validation and test split; then report per-class precision, recall, F1 and a complete 3 by 3 confusion matrix.

5. YOLO status

YOLO is not currently active in the project. There are no labelled bounding boxes, trained YOLO weights or detector runtime in this local app. It would be incorrect to say that YOLO disease detection is already working.

- **Recommended model:** YOLOv8n for detecting a leaf, lesion, pest or damaged region in an image or video frame.
- **Training data required:** real farm images labelled with bounding boxes for leaf, lesion and pest classes.
- **Required files before activation:** a train/validation/test dataset, trained .pt weights, class file, Ultralytics runtime and evaluation metrics such as precision, recall and mAP.
- **Best combined pipeline:** YOLO detects and crops the leaf/lesion, then the MobileNetV2 CNN classifies the disease. Low-confidence results must be escalated to a plant doctor.

6. Safe implementation roadmap

- Collect diverse local images from different farms, seasons and phone cameras.
- Label disease regions and pests with bounding boxes for the YOLO dataset.
- Retrain the CNN with a stratified split and independent test set.
- Integrate CNN and YOLO only after weights, runtime dependencies and reproducible metrics are available.
- Keep medical/agricultural safety notices, low-confidence rules and plant-doctor review in the product.

AgriShield is agricultural decision support. It does not replace laboratory testing, pesticide-label instructions or qualified agricultural advice.